Numerical Electromagnetic Field Analysis for EMC Problems

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(Invited Paper)

***Abstract*—Much progress has been made in the use of computational electromagnetics for the analysis of electromagnetic compatibility (EMC) problems during recent years. This paper reviews the improvements in some of the most important techniques of the field: the method of moments, the finite-difference time-domain method, the finite-element method, the transmission-line matrix method, and the partial-element equivalent-circuit method. The results of computer codes on the basis of such methods have to be validated, and some of the respective possibilities are addressed.**

***Index Terms*—Electromagnetic fields, finite-difference time-domain (FDTD) method, finite-element method (FEM), method of moments (MoM), partial-element equivalent-circuit (PEEC) method, transmission-line matrix (TLM) method, validation.**

I. Introduction

THE LAST ten years have seen an impressive increase in the capabilities of numerical simulation tools and methods for the solution of electromagnetic compatibility (EMC) problems. Tremendous progress in computing power has also taken place. In order to satisfy the increasing demands of industrial practice concerning EMC analysis, it was not sufficient to utilize only the improved computer equipment, rather new computational methods and variants had to be developed.

This paper gives an overview of different numerical methods and their development during recent years. Due to the limited length of this paper, only a selection of the most important methods can be characterized; it is not possible to cover all methods and variants and their combinations.

Typical EMC problems are computation of shielding effectiveness, coupling into lines or cables, analysis of printed circuit boards (PCBs), resonance effects, radiation from antennas, and the investigation of low-frequency magnetic fields. Such problems can cover a broad spectrum of frequencies between hertz and megahertz, and a wide range of dimensions in size between milimeters and hundreds of meters. The number of unknowns to determine the electromagnetic-field distributions may reach hundred thousands of unknowns in the case of integral equation techniques and many millions of unknowns in the case of differential techniques.

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II. Methods

Substantial advancements have been made in enhancing the important numerical techniques, for instance, the method of moments (MoM), the finite-difference time-domain (FDTD) method, the finite-element method (FEM), the transmission-line matrix (TLM) method, and the partial-element equivalent-circuit (PEEC) method. Many numerical methods were invented decades ago. But in all cases, additional novel ideas were required to make them applicable to real-world electromagnetic problems that we have to solve today.

The MoM was introduced into computational electromagnetics by Harrington in the 1960s [1], and it was possible to treat problems in the frequency range up to the first resonances. Since the memory size required increases with number of unknown current amplitudes N according to N^2 and the computation time as N^3, the applicability of the MoM was limited to relatively small structures, measured in wavelengths. However, by using bright matrix reduction techniques the computational costs could be reduced to $N \log N$. The FDTD method has also been specified decades ago. Its broad application to open-space problems was only possible once the perfectly matched layer (PML) technique had been developed by Bérenger [2]. A third example is the TLM, where Johns [3] developed the symmetrically condensed node (SCN) and made TLM an efficient alternative.

There are other methods that have found wide application, for instance, finite integration technique (FIT), multiple multipole (MMP) method, or method of lines (MoL). The most frequently used methods for solving EMC problems, however, are the five methods mentioned above. The methods and new developments in these methods shall be delineated in the following sections.

A. MoM

Integral equation methods have proved to be powerful tools for the numerical computation of electromagnetic fields. For the solution of integral equations, MoM [1] offers a lot of advantages and is frequently used for the analysis of EMC problems. Based on the choice of different basis functions, weighting functions, and discretization procedures, different MoM versions are formulated.

In the following sections, the typical variants of modeling shall be presented, which are able to deal with practical EMC problems. In the case of structures composed of rods or wires like linear antennas, the so-called thin-wire approach is usually applied, which is based on electric-current distributions on the axes of the wires. Another version of MoM uses volume integral

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equations where the unknowns are expressed in terms of volume currents (see, e.g., [4]).

The most frequently used procedure, however, is based on a surface-current formulation. One important aspect in this approach is the appropriate discretization of the surfaces. This can be done by triangular and/or rectangular patches. In the low-frequency range where the dimensions of the structures to be investigated are much smaller than the wavelength, the discretization has to be performed according to geometrical requirements (e.g., considering current and charge variations near corners). In the high-frequency range, the relationship between patch size and wavelength has also to be considered; when using roof-top-type basis functions [5], a maximum patch size of $\lambda/10$ is suitable for smooth-curved structure parts far away from sources or parts of lower interest with respect to a highly accurate field resolution.

In order to allow a simulation for practical configurations with patches that are not planar and have skew angles, a parametric transformation of such curved quadrangular or triangular patches into planar quadratic patches [6] has been introduced for which the conventional basis-function approach is suitable.

The primary formulation of the MoM is an integral equation, using Green's functions in free space. The basis is the operator equation $L(I) = g$ where L is a linear operator, g is the source, and I is the unknown current function. By means of subdividing the structures to be analyzed into N small segments or patches, the current function can be written as

$$I = \sum_{i=1}^{N} \alpha_i I_i \tag{1}$$

with current basis functions I_i and associated unknown coefficients α_i in each segment or on each patch. There are two classes of basis functions I_i: subdomain and entire-domain functions. Among the subdomain basis functions for patches, the Rao–Wilton–Glisson functions [7] have become common. Applying a suitable inner product with testing functions w_m in the range of L, the integral equation is reduced to a system of linear equations written in matrix form

$$[A][\alpha] = [B]. \tag{2}$$

The matrix $[A]$ is fully occupied in most cases. This equation system is solved for the unknown coefficients α by different techniques, e.g., LU decomposition or iterative methods.

The excellence of MoM appears when unbounded radiation problems, and thin wires and homogeneous dielectrics have to be treated.

One of the main goals in recent years has been to extend the MoM to the treatment of complex and electrically large structures. For this purpose, different methods have been developed. One of them is the multilevel fast multipole algorithm (MLFMA) [8], [9], which reduces the complexity for computation time and storage to $O(N \log N)$. Another powerful method to reduce the complexity is the adaptive cross approximation algorithm (ACA) with similar efficiency [10].

An example is given, which demonstrates the efficiency of such approaches. Fig. 1 shows a model of a ship, which is simulated by triangular patches, leading to 95735 unknown current coefficients. The problem is solved by a combination of ACA and singular value decomposition technique (SVD), such that the matrix size is reduced to 5.2 GB and the problem has been solved on a single personal computer (PC).

Fig. 1. Current distribution on a ship, excited by a wire antenna at $f = 120$ MHz on deck. The problem has been solved by MoM using ACA and SVD.

A further possibility is to use hybrid techniques, i.e., to combine the MoM with physical optics (PO) and/or with the uniform geometric theory of diffraction (UTD). Such approaches reduce computer time and memory considerably and enable the investigation of electrically large scatterers [11], [12].

There are specific problems in EMC that require high computational effort, in particular the analysis of shielding effectiveness. In this context, some procedures have been developed which combine the MoM with analytical methods, in particular with the application of 1-D or 2-D wave propagation, and can reach a high numerical precision [13]. Another numerical challenge in EMC is to analyze the electromagnetic radiation of lines or induction in lines close to metallic surfaces, and to determine the voltages coupled into cables, for instance, in aircrafts, ships, or automobiles. In this case, MoM can be combined with TLT and other techniques [14]–[16].

Especially, for structures that are comprised of inhomogeneous materials, a hybrid method consisting of MoM and FEM is suitable. Such methods have already been developed 15–20 years ago [17], [18], and are very useful.

B. FDTD Method

The FDTD method is, in many ways, the pendant to the MoM: FDTD is based on differential equations (Maxwell's curl equations); it is a volume method (the complete computational domain is discretized); it is explicit (there is no necessity to solve a linear system of equations); and it is expressed in the time domain. Due to this, FDTD seems to be mathematically

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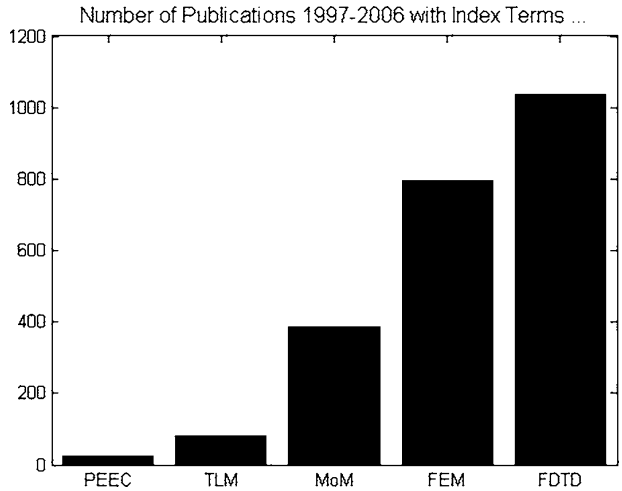

Fig. 2. Number of publications in IEEE periodicals from 1997 to 2006 having index terms "PEEC," "TLM," "MoM," "FEM," and "FDTD," respectively, as found by IEEE Xplore on November 12, 2006.

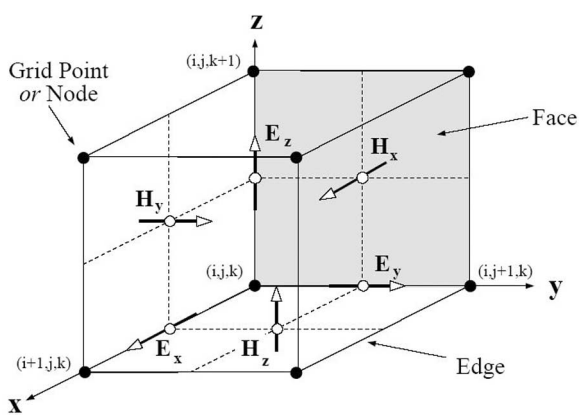

Fig. 3. Basic FDTD grid cell showing the staggering of electric and magnetic field quantities in space.

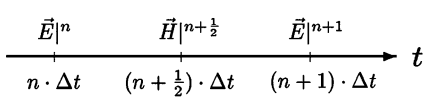

Fig. 4. Staggering of electric- and magnetic-field components in time. The electric field at time $(n+1)\,\Delta t$ is calculated from the electric field at time $n\Delta t$ and the discrete curl of the magnetic field at time $(n+1/2)\,\Delta t$.

less demanding and more intuitive than other numerical methods. Certainly, FDTD is nowadays among the most popular methods, both in the number of available codes (commercial and noncommercial) and the number of scientific publications (Fig. 2).

The FDTD method has its roots in the seminal paper by Yee in 1966 [21]. Although the method of finite differences dates back far beyond (see, e.g., [22]), it was the paper of Yee that first introduced the staggered grids for electric- and magnetic-field quantities in space (Fig. 3) and in time (Fig. 4). Both techniques together ensure a second-order accuracy of the finite difference operators. The whole scheme is numerically stable as long as the time step is not larger than a certain limit (given by the so-called Courant–Friedrichs–Lewy criterion). In cubic grids, it turns out that the time step has to be less or equal to the time that light needs to travel the distance along the space diagonal of the smallest cell in the grid.

In the following decades, several breakthroughs paved the way for the broad application of the FDTD method (for a detailed treatment, the reader is referred to the text books of Kunz and Leubbers [23] and Taflove and Hagness [24]):

- complete understanding of the numerical properties like stability, accuracy, dispersion, and gridding effects [25]–[28];
- development of absorbing boundary conditions culminating in the PML and the uniaxial PML [2], [29], [30];
- adaption to radiation and scattering problems with the total field/scattered field formalism, plane-wave sources, and near-to-far-field transformation [31]–[33];
- adaption to microwave engineering problems with appropriate waveguide sources, lumped elements, and extraction of network parameters [34], [35];
- coupling with thermal solvers, circuit simulators, and semiconductor device solvers [36]–[38];
- modeling of dispersive material properties using auxiliary differential equations and recursive convolution methods [39], [40];
- efficient treatment of oblique, very thin, or very small features using subgrids, subcell models, or surface impedance concepts [41], [42];
- generalization of the Yee grid to nonorthogonal unstructured grids, and adaption to periodic or axial symmetric bodies [43], [44].

Other achievements include pseudospectral techniques, multiresolution algorithms, and hybridization with MoM and FEM [45], [46] with the latter gaining more interest recently.

Today, there seems to be no single class of electromagnetic problems where the FDTD method has not been applied. Radar cross sections, antenna design, signal integrity and power delivery in digital systems, medical imaging, optical devices, grounding and shielding, electromagnetic interference, design of mobile phones [47]—all these issues have been handled successfully. A number of commercial and noncommercial software tools with sophisticated computer-aided design (CAD) interfaces are available to EMC engineers.

FDTD has a natural advantage in the treatment of time-dependent, broadband, nonlinear phenomena and complex inhomogeneous materials. A typical FDTD "experiment" involves the injection of signal (often a Gaussian pulse or a sinusoid) into the computational domain and the recording of (integral) field quantities until all transients have faded away. Subsequent Fourier transform allows the extraction of frequency-domain parameters, e.g., scattering parameters. The ease of time-to-frequency-domain transformation adds to the attractiveness of FDTD.

Situations where FDTD is not a first choice arise when:

- the problem involves mostly empty unbounded spaces;
- the problem shows strong high-Q resonances;
- the problem shows a frequency dependence, which cannot be modeled sufficiently;
- the problem features dictate a time step that is too small to be practical.

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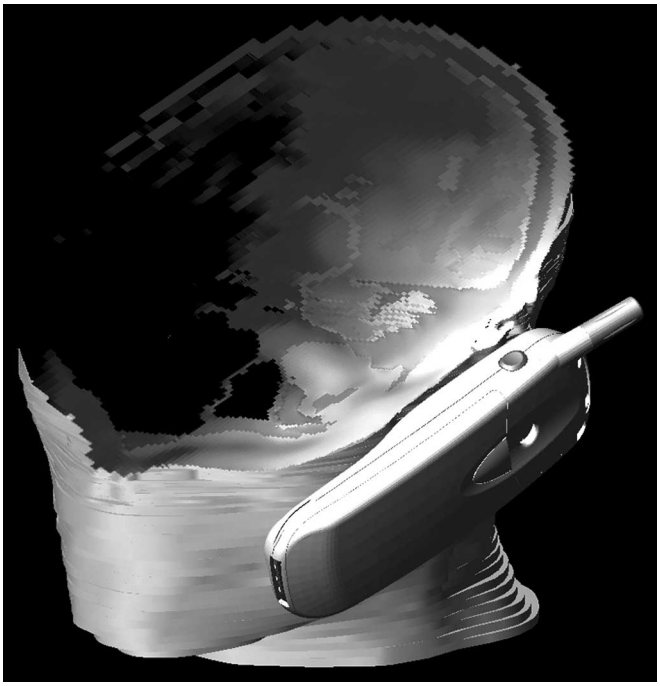

Fig. 5. State-of-the-art FDTD simulation for a cell phone next to a human head using a commercial software tool.[3] The penetration of the electromagnetic field is calculated at 902 MHz. The FDTD grid has a total of 12.5 million cells (including eight layers of PML boundary) with the minimum cell size of the order of 100 μm. CPU time on a 3.4-GHz PC with FDTD hardware acceleration was less than 20 min for over 30 000 time steps.

Workarounds exist, but usually MoM or FEM will be the better choice in these situations, delivering results with less computational effort (it should be noted that a finite-difference frequency-domain has been efficiently used in some of these situations).

Recent trends in FDTD research include unconditionally stable techniques like the alternative direction implicit (ADI) time-marching scheme [48], [49], parallel implementations [50], and the development of dedicated hardware accelerators [51], [52].[1,2] The ADI method overcomes the CFL limit for the maximum stable time step, and allows larger time steps than usual. The other two approaches make full use of today's inexpensive computing power and FDTD's ability to be parallelized in a very efficient manner. Combined, these three efforts will extend the class of FDTD solvable problems far beyond today's tens of millions of grid cells and tens of thousands of time steps (Fig. 5).

C. FEM

FEM is a numerical technique used to solve partial differential equations (PDE). Its origin dates back to Courant [53] in 1943, but it was not applied to electromagnetic problems until the late 1960s [54]. Since then, the method has advanced rapidly in areas such as waveguide problems, microstrips, semiconductor devices, and absorption of electromagnetic radiation in biological bodies. Good introductions to FEM can be found in [55] and [56].

An important feature of the FEM is that it includes the ability to describe the geometry or the media of a given problem with great flexibility. The reason for this is that the geometrical domain of a boundary value problem can be discretized using flexible nonuniform subdomain elements, which is necessary to describe complex geometries. The subdomain elements are called finite elements.

The FEM has a strong resemblance to the MoM because both methods convert either a differential or an integral equation into a matrix equation. As a distinct variation from the MoM, the FEM is based on the physical principle of minimizing the energy of a system [57]. The problem of integrating the PDE is replaced by the equivalent problem of seeking a function that gives a minimum value of a particular integral. Problems of this type are called variational problems. Although both time- and frequency-domain formulations of FEM are known, most implementations have been in the frequency domain.

Let us assume that the problem to be solved is characterized by a partial PDE over the domain Ω

$$L\Phi = f \tag{3}$$

where L is a differential operator, f is the exciting function, and Φ is the physical quantity to be determined. Simple examples are the Laplace equation

$$\nabla^2\Phi = 0$$

or the inhomogeneous Helmholtz wave equation

$$\nabla^2\Phi + k_0^2\Phi = f.$$

On the boundary Γ of Ω, we have boundary conditions of either Dirichlet or Neumann type. The domain Ω is subdivided into finite elements of simple forms with local functions Φ^e. These functions, which are commonly referred to as shape or basis functions, are used to approximate the unknown function within the element. Because of the small variation of Φ within one element, simple functions can be used. Usually, polynomials of first and second degree are used but polynomials of higher degree are possible. In the so-called node-based finite elements, the form of the function in each element is determined by the values at its nodes. Thus, the approximating function can be expressed as a linear combination of basis functions weighted by the nodal coefficients. If the function values Φ_j^e are taken as nodal variables, then the approximating function for element e with n nodes takes the form

$$\tilde{\Phi}^e = \sum_{j=1}^{n} \Phi_j^e \varphi_j^e = \left(\Phi^e\right)^T \left(\varphi^e\right) = \left(\varphi^e\right)^T \left(\Phi^e\right). \tag{4}$$

This equation must be valid for any nodal variable Φ_j^e. Thus, the basis function φ_j^e has to be unity at node j and zero for all other nodes within the element.

One solution method for the FEM is the Ritz or Rayleigh–Ritz method [58], which seeks the stationary point of a variational functional. Under the condition that L is linear, self-adjoint, and positive definite, it can be shown that stationary point of the

[1]Acceleware Corporation, AB, Canada.
[2]EM Photonics, Inc., Newark, DE.
[3]SEMCAD X, Schmid & Partner Engineering AG, Zurich, Switzerland.

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functional

$$F(\tilde{\Phi}) = \frac{1}{2}\langle L\tilde{\Phi}, \tilde{\Phi}\rangle - \langle f, \tilde{\Phi}\rangle \tag{5}$$

is an approximate solution of (3). In other words, F has a minimum at $\tilde{\Phi} = \Phi$, where Φ is a solution of (3).

The inner product over the domain of two functions is defined as

$$\langle a, b\rangle = \int_{\Omega} a \cdot b \cdot d\Omega$$

or for vector functions

$$\langle \vec{a}, \vec{b}\rangle = \int_{\Omega} \vec{a} \cdot \vec{b} \cdot d\Omega.$$

The choice of the latter extends the validity of the variational expressions to vector fields. Inserting (4) into (5) provides [59]

$$F(\tilde{\Phi}) = \sum_{e=1}^{M} F^e(\tilde{\Phi}^e)$$

where M is the total number of finite elements. The local functional can be written as

$$F^e(\tilde{\Phi}^e) = \frac{1}{2}(\Phi^e)^T \int_{\Omega^e} (\varphi^e) L (\varphi^e)^T d\Omega (\Phi^e) - (\Phi^e)^T \int_{\Omega^e} f(\varphi^e) d\Omega.$$

In matrix notation

$$F^e(\tilde{\Phi}^e) = \frac{1}{2}(\Phi^e)^T [K^e](\Phi^e) - (\Phi^e)^T (b^e) \tag{6}$$

where $[K^e]$ is an $n \times n$ matrix with coefficients according to

$$K^e_{ij} = \int_{\Omega^e} \varphi^e_i L \varphi^e_j \, d\Omega$$

and

$$b^e_i = \int_{\Omega^e} f \varphi^e_i \, d\Omega. \tag{7}$$

Considering all elements and introducing a global node numbering, (6) can be expressed as

$$F = \frac{1}{2}(\Phi)^T [K](\Phi) - (\Phi)^T (b).$$

All contributions, which belong to a single node, are added up so that each node appears only one time. (Φ) is an $N \times 1$ vector where N is the number of basis functions with unknown expansion coefficients. The vector (b) describes the excitation of the system with values according to (7).

To minimize the functional, it is partially differentiated with respect to the unknown coefficients and set to zero. From this, N equations are obtained that can be written as a matrix equation

$$[K](\Phi) = (b).$$

This equation system is very sparse and typically symmetric. Its nonzero entries provide the relationship among field or voltage of adjacent elements. It can be solved by a direct or iterative solver.

Throughout the 1970s and 1980s, the computation of electromagnetic problems with the FEM suffered from "spurious modes" [60], which are numerically valid but physically incorrect solutions because node-based elements were employed to represent vector electric and magnetic fields. A way out of the dilemma was the introduction of edge-based vector basis functions [18], [61], [62] where the tangential continuity of the field across the boundaries between adjacent finite elements is guaranteed. Here, the unknowns are associated with element edges. These types of functions are now extensively applied in electromagnetic FEM computations. Edge-based functions were already described by Whitney almost four decades ago [63]. They started appearing in the computational electromagnetics literature in the late 1980s, as a consequence of the famous paper by Nedelec [64].

Exterior or open problems such as radiation and scattering present a challenge to the finite element technique. Covering the entire space with finite elements would require infinite computer resources. Therefore, the solution domain has to be truncated to a finite size by using a domain termination boundary. Several approaches have been proposed, all of which have their strengths and weaknesses. Among them, the PML [53] has shown considerable promise in dealing with exterior problems. It is important to mention the finite element-boundary integral (FE-BI) method for the 3-D implementation of the FEM in this context. This technique is a hybridization of the traditional MoM with the FEM for open problems [65], [66].

D. TLM Method

In the beginning of the 1970s, Johns and Beurle [67] developed the TLM method, which has also become an established technique for the numerical computation of 3-D electromagnetic fields. It is based on the analogy of electromagnetic wave propagation and the propagation of impulses in transmission line networks [68], [69]. In TLM, this analogy is further developed to construct 3-D networks where voltages and currents can model electric and magnetic fields [70]. The basic structure used for this purpose is the SCN, which has originally been proposed by Johns [71] and is shown in Fig. 6. Most TLM models operate in the time domain, although a frequency-domain formulation is known [72]. Compared to the popular FDTD method as described in Section II-B, the TLM has a larger computational effort [73].

A single SCN models the propagation in a small block of space, the size of which should be typically less than one tenth of the wavelength of the highest frequency of interest. As illustrated in Fig. 6, a node is formed by six interconnected transmission lines, where each line is characterized by two ports in each coordinate direction. At each time step, 12 voltage pulses V^i are incident on each SCN. After scattering, 12 reflected pulses are generated according to

$$(V)^r = [S](V)^i \tag{8}$$

with $[S]$ as the scattering matrix. The reflected pulses become incident into the corresponding ports of the neighboring nodes

$$(V)^i_{k+1} = [C](V)_k \tag{9}$$

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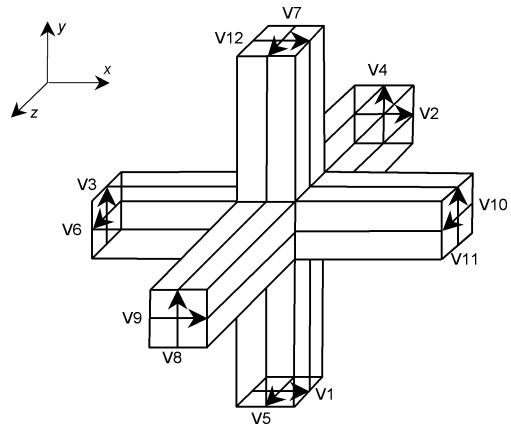

Fig. 6. Symmetrical condensed node (SCN). All TLM nodes that were developed before the SCN could have been derived from equivalent circuits, and a simulation in a laboratory by a grid of lines were theoretically possible [75]. This is not true for the SCN, which is a mathematical construct. The scattering process takes place at the node center.

where $[C]$ is the connection matrix. The TLM method is, thus, the repetition of steps (8) and (9) as long as transient system responses are decayed or have provided the desired information. The presence of dielectric or magnetic material can be taken into account by adding six extra ports to the SCN according to Fig. 6 [74]. Absorbing boundary conditions have been implemented to extend the method [74].

Various authors have suggested formal derivatives of the basic SCN in order to seek an optimum numerical behavior; worth mention are the general SCN [76], the supercondensed node [77], and the hybrid node [78]. In order to enhance the flexibility for complicated structure shapes, multigrid schemes are available [79], [80], and customized thin-wire thin-wall approaches have been included [81], [82]. To save memory resources, the alternating transmission line matrix (ATLM) scheme was suggested, later improved by the alternating rotated TLM (ARTLM) [83], [84]. Over the years, TLM has been applied to all aspects of EMC.

E. PEEC Method

The PEEC method was introduced by Ruehli in 1974 [85] mainly for the analysis of interconnect and packaging structures. The formulation is derived from an integral equation and provides a full wave solution to Maxwell's equations. The analysis applies to both time and frequency domain. A major difference from the MoM is that the problem under consideration is transformed into the circuit domain where circuit analysis techniques can be applied. Despite the fact that the PEEC method has an exact field theoretical basis, it was not primarily developed for the computation of electromagnetic fields. Instead, the circuit designer has a tool at hand to analyze parasitic effects in connecting structures of circuits without leaving the familiar area of network theory [86]. A detailed description of the approach can be found in [87]. This method was strongly developed further in the 1990s where retardation, external-field excitation, and the treatment of dielectric material were investigated [88]–[90].

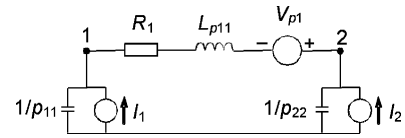

Fig. 7. Equivalent circuit model of a PEEC cell.

The first step in obtaining a solution using the PEEC method requires the subdivision of the structure in geometrically and electrically small cells where rectangular current carrying inductive volume cells, charge carrying surface cells, and dielectric volume cells are distinguished. Both charge and current basis functions are taken to be constant over each cell. It has to be noticed that nonorthogonal elements are also known [91], and more recently, a surface-based formulation of the PEEC method has been published [92]. The so-called Galerkin's weighting procedure is applied to enforce the integral equation

$$\begin{aligned}\vec{E}^i(\vec{r},t) = \frac{1}{\kappa}\vec{J}(\vec{r},t) &+ \mu\int_{V'} G(\vec{r},\vec{r}\,')\frac{\partial\vec{J}(\vec{r}\,',t-\tau)}{\partial t}dV' \\ &+ \mu\varepsilon_0(\varepsilon_r-1)\int_{V'} G(\vec{r},\vec{r}\,')\frac{\partial^2\vec{E}(\vec{r}\,',t-\tau)}{\partial^2 t}dV' \\ &+ \frac{1}{\varepsilon_0}\nabla\int_{V'} G(\vec{r},\vec{r}\,')\rho_t(\vec{r}\,',t-\tau)dV'\end{aligned}$$

at the cells [93] where G is Green's function of free space, and J and ρ are, respectively, the current and charge densities at the source cells. From the resulting expression, an equivalent circuit for a PEEC volume cell and two surface cells can be derived, forming the building block of the total PEEC model, shown in Fig. 7, taken from [93]. In Fig. 7, L_{p11} is the partial inductance, R_1 is the series resistance of the cell, and p_{11} and p_{22} are the potential coefficients.

Application of the continuity equation to all N network nodes of the structure connects the nodes and provides N node equations for the computation of the overall circuit [94]. The resulting equation system can be solved by appropriate network solvers. In the frequency domain, the modified nodal analysis (MNA) [95] can be applied. Various papers deal with the problem of stability in both time domain ("late-time instabilities") and frequency domain [96], [97]. It seems that the solution of the stability problem still remains challenging, especially in the time domain.

III. Validation

Once the numerical solution of the electromagnetic problem has been achieved, the question arises whether the results are convergent, numerically stable, and reliable. Do the results conform to the physical requirements and are they related to the real situation? All results have to be validated, even if the code used is established and trustworthy. A new IEEE standard [98] will treat this subject. Essentially, two different methods can be distinguished to determine the numerical credibility of the computed results [19], [20], the so-called internal and external checks.

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A. Internal Checks

The validation by means of internal checks is performed by the code itself. A first imperative, before results are computed, is to check the discretized model, which is—at least in case of a complex structure—a simplified model. One of the first questions should be whether the discretization has really been performed according to the geometrical requirements, or in the high-frequency range, according to the necessary relationship between the patch size and wavelength. A closer look at the volume or surface grid can often reveal deficiencies of the discretization effort.

The most important internal checks comprise the power budget and the verification of the boundary conditions. In case of radiation problems, the power budget requires that the input power of an antenna must be equal to the dissipated power plus the radiated power P_{rad}, which can be checked in the far field, for instance, by integrating over a fictitious far-field sphere S

$$P_{\mathrm{rad}} = \frac{1}{2} Re \oiint_S (\hat{\vec{E}} \times \hat{\vec{H}}^*) d\vec{S}.$$

In case of an MoM simulation, it is a good idea to check the surface current distribution, which should be physically reasonable. For instance, at high frequencies it should show patterns according to the wavelength. When interpreting numerical results, it makes little sense to consider only a single value of a physical quantity like I or E. In case of a single frequency or a single time step, one should observe the spatial distribution of the currents and/or fields and choose at least 8–12 vectors per wavelength. Animations of these distributions are sometimes very helpful.

The boundary conditions can be checked by some simple well-known principles. For ideally conducting parts, the condition $E_{\tan} = 0$ can be inspected, which means that only normal components of E can appear. Vice versa, only tangential components of H should then occur. At interfaces between dielectric bodies, the tangential components of E and H should be continuous.

B. External Checks

Performing only internal checks like power budget or boundary conditions could lead to an unsatisfactory level of credibility. Therefore, external checks are also recommended.

Often simple similar structures, which provide an analytical solution, can be used for this purpose, for instance, a dielectric sphere instead of a human head if the specific absorption rate is to be investigated. Another example is a patch antenna where an analytical solution exists for ideally conducting plates with dielectric material in between. In this case, one could compare the input impedances of numerical and analytical computations as a function of frequency.

In case of electrically small wire antennas, the radiation resistance can be approximated by simple formulae, which are available for rod and loop antennas. The imaginary part of a monopole antenna should behave like $1/\omega C$ with

$$C = \frac{\varepsilon 2\pi l}{\ln \frac{l}{\sqrt{3}R}}$$

and l as length of the rod, and R as its radius. If such an antenna is placed on a confined metal structure (ship, aircraft), the real part should notably increase due to the radiation of this structure.

Sometimes, it is possible to check a simulation result at a very low frequency, e.g., in case of a TEM cell. The generator should then see the lumped resistor at the end, and the current can simply be calculated by Ohm's law. It is not a sufficient condition but a *sine qua non*.

One of the most satisfying validations is the comparison with measured data. Insufficient correlation can be caused not only by simulation errors (modeling or computational errors as stated above, or infeasibility of the code applied), but also by measuring errors.

If no analytical solution for a similar model exists, one can compare the numerical results with those obtained with alternate numerical techniques, e.g., codes based on FDTD or FEM instead of MoM, or a code on the basis of TLM instead of FDTD, if such a different code is available.

IV. Outlook and Conclusion

The development of numerical techniques for electromagnetic field analysis has matured considerably over the last ten and more years. EMC engineers can now choose between integral or differential, surface or volume, time domain or frequency domain, commercial or open-source tools of any level of sophistication. This "commoditization" of numerical electromagnetic field analysis entails obviously many benefits for the user. Simulations can be done many times easier, faster, cheaper, and more accurate than ten years ago.

Where will future research in this field be headed? The authors of this paper see among others the following trends:

- extension of the range of applications (from very small scales to very large scales, different fields of interest, etc.);
- interfacing of electromagnetic field simulation tools with other solvers (heat conduction, device physics, etc.);
- refinement of existing techniques (faster speed, parallelization, etc.);
- adaption of hardware design to software tool requirements.

The question if a fundamentally new and technically convincing numerical method will be invented some time remains open.

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